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Statement of Authorship

I hereby declare that this document has been composed by myself and describes my own work, unless otherwise acknowledged in the text.

Passau, August 19, 2025

Abstract

A short summary of what is going on here.

Deutsche Zusammenfassung

Kurze Inhaltsangabe auf deutsch.

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1. Introduction

This chapter should contain

1. A short description of the thesis topic and its background.
2. An overview of related work in this field.
3. Contributions of the thesis.
4. Outline of the thesis.

2. Preliminaries

Before tackling the problem, we will introduce some concepts that are necessary for the understanding of this problem.

Definition 2.1 (Graph Embedding). *a Graph embedding of a graph G on a*

Definition 2.2 (Level Assignment). *A level assignment l of a level graph G is a mapping of vertices of G to levels.*

$$l : V(G) \rightarrow \mathbb{N}$$

Definition 2.3 (Proper Level Graph). *a Proper Level Graph G is a graph with a level assignment l that defines in which level every vertex is while respecting that for every edge $(u, v) \in E(G)$ $|l(u) - l(v)| = 1$*

Definition 2.4 (Planar Graph). *a Planar Graph G is a graph that can have an embedding on a 2-dimensional Euclidean space $\mathcal{R}$ in a way that edges only intersect with each other at their endpoints.*

2.0.1. Equivalence relation and the 2-SAT formulation

Randerath et al. [RSB⁺01] present some simple constraints that decide the existence of a level-planar drawing. The constraints are as follows:

- *consistency:* $uw \iff \neg wu \quad \forall uw \in \mathcal{V}$
- *transitivity:* $(uw \wedge wx) \Rightarrow ux \quad \forall uw, wx \in \mathcal{V} \text{ and with } ux \in \mathcal{V}$
- *planarity:* $uw \iff vx \quad \forall uw, vx \in \mathcal{V} \text{ with } (u, v), (w, x) \in E \text{ and are independent}$

Definition 2.5 (Equivalence relation). *An equivalence relation on a set X is a binary relation $\sim$ on X satisfying the following constraints:*

- *reflexivity:* $a \sim a \quad \forall a \in X$.
- *symmetry:* $a \sim b \text{ implies } b \sim a \quad \forall a, b \in X$.
- *transitivity:* $a \sim b \wedge b \sim c \text{ implies } a \sim c \quad \forall a, b, c \in X$.

The planarity constraint can be formulated as an *equivalence relation* over the set $\mathcal{V}$ according to Randerath et al. [RSB⁺01].

Definition 2.6 (Orbit relation).

Let's consider a proper level graph $G = (V, E)$, and a boolean set $\mathcal{V} = \{uw \mid u, w \in V, l(u) = l(w)\}$. The orbit relation is an equivalence relation on the boolean set $\mathcal{V}$ with a binary relation $\sim$ on $\mathcal{V}$ satisfying the following constraints:

- *reflexivity*: $uw \sim uw \quad \forall uw \in \mathcal{V}$
- *symmetry*: $uw \sim vx$ implies $vx \sim uw \quad \forall uw, vx \in \mathcal{V}$
- *transitivity*: $(uw \sim zy) \wedge (zy \sim vx)$ implies $uw \sim vx \quad \forall uw, zy, vx \in \mathcal{V}$

The interpretation of the orbit of a variable $uw \in \mathcal{V}$ is that for each boolean variable $vx \in \mathcal{V}$ in the orbit of uw we have: $(u < w) \iff (v < x)$. [RSB⁺01]

Theorem 2.7. A proper level graph G is level planar iff there exists no $uw \in \mathcal{V}$ such that wu is in orbit of uw .

Proof. Refer to Theorem 6 from [RSB⁺01]. □

by considering the orbit which represents the *planarity* constraint and by considering the *consistency* constraint, that is necessary to prove the existence of a level planar based on the Theorem 2.7. Randert et al [RSB⁺01] proves that the transitivity clause is not needed for the satisfiability of this system of constraints and it is equivalent to the satisfiability of a reduced constraint system which omits the transitivity clause.

The reduced constraint system can be expressed in terms of 2-SAT, which can be solved efficiently with a quadratic-time algorithm for testing level planarity [Kro67]. Randert et al. also prove that a solution can be made transitive without invalidating the other constraints in Theorem 2.7.

2.0.2. Hanani-Tutte drawing and Embedding

Theorem 2.8. If G has an $(x-)$ monotone drawing with every pair of independent edges crosses evenly then G has $(x-)$ monotone embedding with the same vertex locations (over x).

Proof. Refer to theorem 3.1 from [FPSŠ12] □

2.0.3. Equivalence classes of the orbits and Hanani-Tutte drawing

In this section we want to link the equivalence classes with the Hanani-Tutte drawing induced from the truth assignment of the 2-SAT formulation that is based on equivalence classes of the orbits.

Definition 2.9 (Equivalence class). For a given set X and an equivalent relation $\sim$ in X , for an element x of X , the equivalence class $[x]_\sim = \{y \sim x, y \in X\}$.

The set of all equivalence classes of X with a relation $\sim$ is represented with $X / \sim$

Definition 2.10 (Orbit class). For a boolean variable $uw \in \mathcal{V}$, the orbit class $[uw]_\sim$ is the set of all elements $vx \in \mathcal{V}$ with $vx \sim uw$.

The set of all orbit classes of $\mathcal{V}$ with the relation $\sim$ is $\mathcal{V} / \sim$.

Lemma 2.11. $\forall uw \in \mathcal{V} \forall vx \in [uw]_{\sim} : \neg vx \in [\neg uw]_{\sim}$

Proof. by the consistency clause, $\forall uw \in \mathcal{V} : uw \iff \neg wu$ and $\forall vx \in [uw]_{\sim} vx \iff uw$ therefore

$$\forall uw \in \mathcal{V} \forall vx \in [uw]_{\sim} vx \iff \neg wu$$

thus

$$\forall uw \in \mathcal{V} \forall vx \in [uw]_{\sim} \neg vx \iff wu \iff \neg uw$$

and that's why

$$\forall uw \in \mathcal{V} \forall vx \in [uw]_{\sim} \neg vx \in [\neg uw]_{\sim}$$

□

Each assignment of the equivalence classes of the orbit is represented in the hanani-tutte drawing Γ^* of G^* in the level l where the single vertex v' of origin v in G is one of the pair of vertices in the equivalence class and the other vertex w of the pair is the edge e' passing through the level l with an endpoint of origin w in the graph G . If for example we assign one to $[vw]_{\sim}$, then the vertex v' lays before e' on the level l in the drawing Γ^* .

Definition 2.12 (Horizontal Gang). *a Horizontal Gang is a set of distinct vertices that belong to the same level of the level graph G and that are the distinct endpoints of adjacent edges.*

Definition 2.13 (Base Orbit Class). *a base orbit class $\mathcal{B}([uv]_{\sim})$ is the merge of $[uv]_{\sim}$ and $[vu]_{\sim}$. The orbit classes $[uv]_{\sim}$ and $[vu]_{\sim}$ are the directions of the base orbit class $\mathcal{B}([uv]_{\sim})$.*

Lemma 2.14. *Let a level planar graph $G = (V, E)$ with a level assignment l and let the drawing Γ^* be the induced Hanani-Tutte drawing of the graph G^* induced from the orbit classes of G . An odd crossing in Γ^* is equivalent to a cyclic relation between a triple of members of a horizontal gang V_{HG} , in other words a pair or triple of base orbit classes of members of the horizontal gang can have directions that are cyclic.*

Proof. TODO proof

□

The algorithm that we designed might merges some of the orbit classes, we want to separate the orbit classes before the execution of the algorithm from any other state of the $\mathcal{V}/\sim$, for this reason we introduce this definition.

Definition 2.15 (first-gen orbit class). *A first-gen orbit class is an orbit class purely based on the planarity constraint, wich is not contaminated with other elements of $\mathcal{V}$ that are not par of it from the planarity constraint.*

3. Transitivity constraint and Merging of the equivalent classes

After linking the Equivalent Classes to the hanani-tutte drawing induced by the 2-Sat formula (insert paper), and also presenting the cases that lead to a cyclic relation between some vertices of the same level, we can now present our algorithm that reduce the set equivalent classes resulted from resolving the 2-SAT formula (insert papers).

3.1. Algorithm description

We have as input a planar proper level graph $G = (V, E, l)$ with a level assignment l , and equivalence classes based on the 2-SAT formulation of level planar embedding presented by Randert et al [RSB⁺01]. The 2-SAT formulation tests whether a proper level graph is planar. The truth assignments that solve the 2-SAT formula give embeddings that respect the planarity constraints but some of the assignments can't be translated to an embeddings in a plane surface due to the existence of cyclic relation between some of the vertices from the same level. These set of these assignments is equal to the set of possible combinations of assignment of the equivalence classes.

The goal of the algorithm is to merge some of the equivalence classes which means less possible truth assignment claiming that with the newly merged equivalence classes all the possible assignments of these equivalence classes lead to an embedding.

We first make the edges of G point to the same direction.(part of the computation of the equivalent classes)

We traverse the graph level by level, we also keep track of the vertices that we went through by adding them to a graph G' that is initially an empty graph. Let h be the current level. In each level we go through all the vertices, for each vertex v we execute two main tasks. The first task is to merge the equivalence classes of neighboring vertices to v that belong in the same adjacent level to level h . The second task is to merge the equivalence classes of pair of vertices (u, w) with u and w are in different connected components C_1 and C_2 of $G - \{v\}$ and v is a cut vertex of G connecting C_1 and C_2 .

For the first task, we consider vertices in $V_v^{(h-1)}$ adjacent to the vertex v on the level $h - 1$. We compute an ordering $\mathcal{V}$ of the vertices in $V_v^{(h-1)}$ so that for all pairs of vertices $(u, w) \in V_v^{(h-1)} \times V_v^{(h-1)}$ and $u < w$ the equivalence classes point to the same direction.

Algorithm 3.1: MERGEADJACENTEDGESTORBITCLASSES

Input: proper level Graph $G = (V, E, l)$ and Eq
Output: Eq
 /* reducing the orbit classes of adjacent edges */

```

1 forall  $h \in l(V)$  do
2   forall  $v \in \{V \mid l(v) = l(h)\}$  do
3     visited  $\leftarrow$  null
4     parentEq  $\leftarrow$  null
5     reverse  $\leftarrow$  false
6     forall  $w \in \text{adj}(h, v)$  do
7       member  $\leftarrow$  visited
8       while member.next  $\neq$  null do
9         if parentEq = null then
10          parentEq  $\leftarrow$  eq( $u, w$ )
11         if  $((w, u) \in \text{parentEq})$  then
12          reverse  $\leftarrow$  true
13          member.previous  $\leftarrow$ 
            {data:  $w$ , perivous: member.previous, next: member}
14         if  $(\text{eq}(u, w) \notin \text{parentEq}) \wedge (\text{eq}(w, u) \notin \text{parentEq})$  then
15           if reverse then
16             Merge eq( $w, u$ ) with parentEq
17           else
18             Merge eq( $u, w$ ) with parentEq
19         member.next  $\leftarrow$  {data:  $w$ , perivous: member, next: null}
20       if visited = null then
21         visited  $\leftarrow$  {data:  $w$ , perivous: null, next: null}
```

We merge those equivalence classes of the pairs $(u, w) \in \mathcal{V}$ with $u < w$. We merge also the equivalence classes of the pairs $(u, w) \in \mathcal{V}$ with $u > w$ to hold the consistency clause.

We will also do the same steps for vertices in $V_v^{(h+1)}$ that are adjacent to the vertex v on the level $h + 1$.

For the second task. we add v to G' . If the vertex v is a cut-vertex in G' , we look for the connected components S_C in $G' - \{v\}$. We pick an arbitrary absolute ordering $\mathcal{V}_{S_C}$ of the connected components in S_C . For each pair of connected components $(C_1, C_2) \in \mathcal{V}_{S_C}$, for each $w_1 \in C_1$ with $l(w_1) = h - 1$ we merge the equivalence classes of w_1 with all $w_2 \in C_2$ and $l(w_2) = h - 1$.

3.2. Correctness of the algorithm

3.2.1. Merging orbit classes of horizontal gangs

In this section, we will prove that the part of algorithm that might merges some of the orbit classes between members of the same horizontal gangs, and that leads to a new set of orbit classes that for all their assignments there exists a weak Hanani-Tutte drawing in G^* .

Lemma 3.1. *let $(w_1, \dots, w_j)$ subset HG_A of vertices of a horizontal gang HG with no assignment of the orbit classes in HG_A resulting in a cyclic relation. let $w_{j+1} \in HG$ After*

the algorithm 3.1 treats w_{j+1} in the loop. $HG_A \cup \{w_{j+1}\}$ has no assignement of the orbit classes in $HG_A \cup \{w_{j+1}\}$ that lead to a cyclic relation between some of the members of the subset $HG_A \cup \{w_{j+1}\}$

Proof. We proceed the proof by induction on members of the horizontal gang HG_{i+1} which correspondence to the loop of the gang in the algorithm.

The base case consists only from a single vertex member of the horizontal gang HG_{i+1}^1 which holds the invariant of the induction on the horizontal gangs and also hold the invariant of having acyclic relations between the members of the horizontal gang HG_{i+1}^1 since the gang has no cyclic relations, since there are no relations.

For the induction step, let's consider that the provisional horizontal gang HG_{i+1}^j consists of $(w_1, \dots, w_j)$ members, with $j < l$, we want to prove adding the vertex member w_{j+1} to the gang HG_{i+1}^j will hold both invariants.

To prove this we will prove by induction that merging the equivalence class between w_{j+1} and the members of HG_{i+1}^j hold the two invariants. Let VHG the set of visited members of the horizontal gang HG_{i+1}^j .

For the base case we have only one visited member $VHG_1 = \{w_1\}$ from the gang HG_{i+1}^j . Which means we are considering only one equivalence class $eq(w_1, w_{j+1})$, which doesn't lead to any cyclic relation since we need a three vertices at least for that to be possible.

For the induction step, let $VHG_k = (w_1, \dots, w_k)$ with $k < j$. The equivalence class let's consider the $eq(w_k, w_{j+1})$. If the equivalence class origin is different than the root equivalence class origin then we merge $eq(w_k, w_{j+1})$ with the root equivalence class $eq(w_0, w_1)$.

This first keeps the first invariance (TODO give a name to the invariance?) because merging $eq(w_k, w_{j+1})$ and the root equivalence class, without losing generality, is only possible if the opposite direction of $eq(w_k, w_{j+1})$ which is $eq(w_{j+1}, w_k)$ doesn't exist in the parent equivalence class, which means the consistency clause is still respected, and since we are just merging equivalence classes and not adding any new relations between vertices the planarity constraint is also respected. For the second invariant (TODO give it a name), let's consider prove by contradiction that merging the origin equivalence class of w_k and w_{j+1} , with a direction depending on the phase of the algorithm, with the parent equivalence class will lead to a cyclic relation in one of the 'new' triples (x, w_k, w_{j+1}) with $x \in VHG_k - \{w_k\}$. For the first mode, where $inverse = false$ which means that we didn't find a member of the gang that must be after w_{j+1} . 'That can't happen if the base of equivalence class (w_k, w_{j+1}) is not part of the base of the parent equivalence class, because whatever the direction of the base of the equivalence class (w_k, w_{j+1}) , it doesn't lead to a cyclic relation if we merge the origin of that equivalence with one of the direction with the parent equivalent class. For the case where we only have the parent equivalence class between the triple. It can't be an original equivalence class as these don't lead to a cyclic relation (TODO lemma?). If the equivalence class in the triple is a result of one or multiple merges, We know that a triple of gang members that can have a cyclic relation don't share more than one edge with any other triple of gang members (lemma) which means that, we can't have this case since if we had at least one merge of the equivalence class in the triple it means at least two equivalence classes where part of another triple of gang members. For the second mode, where $inverse = true$ where we found that w_{j+1} needs to be before a gang member of VHG . For each triple (x, w_k, w_{j+1}) with $x \in VHG_k - \{w_k\}$, they can induce two equivalence classes or one. In the case of two, the base equivalence class between x, w_{j+1} is of the direction from w_{j+1} to x and the base equivalence class x, w_k is of a direction from x to w_k since x comes before w_k in visited linked list. Since we are in the reverse mode of the algorithm, the algorithm will merge the base equivalence class of w_k, w_{j+1} with the

direction from w_{j+1} to w_k with the other mentioned induced equivalence classes, which contradicts. For the case, where we have only one equivalence class induced from the triple, it's not possible due to the same contradiction from the first mode.

The merge of the appropriate equivalence class from the base of (w_{j+1}, w_k) doesn't affect any existing triple order, since we are just changing the new equivalence class.

Therefore merging the equivalence classes between w_{j+1} and the members of HG_{i+1}^j , when needed, hold the two invariants.

Thus adding the vertex member w_{j+1} to the gang HG_{i+1}^j will hold both invariants.

□

Theorem 3.2. *The algorithm 3.1 is correct and hold the invariant H_1 while removing all the possible cyclic relations between members of the same horizontal gang.*

Proof. We can proof the correctness of this algorithm 3.1 by proving that the resulted orbit classes of the algorithm have no assignment that for a triple or pair of base orbit classes $\mathcal{B}(\mathcal{V}_{HG})$ of members of a horizontal gang we can find a combinaison of directions that lead to a cyclic relation between orbit classes from the triple or pair of base orbit classes $\mathcal{B}(\mathcal{V}_{HG})$. By Theorem 2.8, we can prove it by induction on odd pairs of the Hanani-Tutte drawing induced by a truth assignement. And by Lemma 2.14 We can prove it by induction on the horizontal gangs and removing any possible combinaison of directions between base orbits of members of the horizontal gang.

We want also to hold an invariant H_1 of having the orbit classes not breaking the *consistency constraint* and the *planarity constraint* which also mean that any combination assignements of the orbit classes can be drawn as a hanani-tutte drawing.

The base case is our *first-gen* orbit classes of the graph G induced from the 2-SAT formulation. all the possible combinations of assignements of the first-gen orbit classes can be lead to a hanani-tutte drawings [BRS22].

For the induction step, let's consider that the possible combinations of assignements of the set of the orbit classes Or_i of graph G after treating i sets of horizontal gangs, holds the invariant H_1

We want to prove that by merging some orbit classes corresponding to the horizontal gang HG_{i+1} resulting in the set of equivalence classes Eq_{i+1} the invariant H_1 holds and also taht we removed any possilbe cyclic relation between any triple of the horizontal gang HG_{i+1} .

let's consider $(w_1, \dots, w_l)$ the vertices of the horizontal gang HG_{i+1} .

to prove that the merging done in the algorithm is correct, we need to prove that the merging resulting in Eq_{i+1} holds the invariant of H_1 and also hold the invariant H_2 of having acyclic relations between the members of the horizontal gang HG_{i+1} .

Which means that treating the horizontal gang HG_{i+1} holds the invariant and also removes any possible cyclic relation between the horizontal gang memberes of HG_{i+1} from the set of resulted equivalence classes.

Therefore, we proved that the algorithm removes all the cyclic relations between any members of the same horizontal gang. Which is all what we need to prove since the proof of transforming strong hanani-tutte drawing to weak hanai-tutte drawing from theorem 3.1 from .. induce on the the number of odd pairs between adjacent edges. The gang members are only the vertices that can lead to the odd crossings described in theorem 3.1. □

For the induction step, let's consider a subgraph G_i of our graph G . Let v_{i+1} be the vertex to add to the subgraph G_i .

If v_{i+1} has no more than one adjacent edge

The algorithm is based on the fact that we can convert a 2-SAT truth assignment to a strong hanani-tutte drawing and that we can convert a strong hanani-tutte drawing to a planar drawing.

The assignment of equivalent classes represent the set of solutions from solving the 2-SAT assignment.

The algorithm reduce the set of truth assignment solutions to get rid of false positives and that by reducing the number of equivalent classes by concatenating some of them, which lead to smaller number of possible solution that can lead to a level planar drawing.

Proof. Since we are just reducing the set of truth assignment that already satisfies 2-SAT, the planarity and the consistency clauses are still respected.

if we take a truth assignment ϕ from 2-sat solutions, we can induce a strong hanani-tutte drawing. The goal is to convert this strong hanani-tutte drawing into a x-monotone embedding which is a planar drawing.

To convert the strong hanani-tutte drawing to weak hanani-tutte, we must remove any odd crossing of adjacent edges by making it an even crossing. We either have two edges consecutive in the rotation of the shared endpoint, so we can just cross them one more time near the shared endpoint, this doesn't change the order of the vertices in the hanani-tutte drawing Figure 3.1. The other cases are when there is at least an edge between the two edges that crosses oddly and this edge crosses evenly all edges of the constructed graph so far, we have three major cases (TODO describing the cases from the hanani-tutte to xmonotone paper) , and we can get rid the odd crossing by concatenating the equivalent classes while still respecting the consistency clause Figure 3.2

After the modification, each element of the set of solutions from the reduced equivalent classes will lead to a weak hanani-tutte drawing, there is a special case in weak hanani-tutte that doesn't lead to an equivalent x-monotone embedding wich means that we need to change the relative order of some vertices. We mimick the construction in the proof of theorem 2.1 (reference hanani-tutte to xmonontone paper) of the graph G' (which equals G at the end of the construction) , when the algorithm reach the special case which is when the new added x_1 adds more than one edge to the graph G' and it is not connected to the last added vertex before it in G' , from the proof we can notice that we fix the order of two connected comps to each other if they don't have any relative order (TODO explain what's a relative order in this context), if they do, meaning that one encapsulate the other then each vertex of the encapsulating component forces that it has the same equivalent class to all vertices of the encapsulated comp. From the algorithm we take an arbitrary order of the components (C_1, C_2) and we force all elements of C_2 to have the same equivalent class with the same vertex from C_1 . If C_1 is the encapsulated component, we just switch C_1 with C_2 .

□

Algorithm 3.2: EQUIVALENCE CLASSES REDUCTION

Input: proper level Graph $G = (V, E, l)$ and Eq

Output: Eq

```

1  $\mathcal{V} = \{uw \mid u, w \in V, u \neq w, l(u) = l(w)\}$ 
  /* reducing the equivalence classes of adjacent edges */
2 forall  $h \in l(V)$  do
  /* i still need to make sure adding  $h + 1$  is valid here. */
3  forall  $v \in \{V \mid l(v) \in \{h - 1, h + 1\}\}$  do
4     $S \leftarrow \emptyset$ 
5     $\mathcal{V}_{\text{ord}} \leftarrow \emptyset$ 
6    forall  $uw \in \mathcal{V}_{\text{adj}}(h, v)$  do
7      if  $wu \in S$  then
8         $\mathcal{V}_{\text{ord}} \leftarrow \mathcal{V}_{\text{ord}} \cup \{wu\}$ 
9         $S \leftarrow S \cup \text{Eq}(wu)$ 
10     else
11        $\mathcal{V}_{\text{ord}} \leftarrow \mathcal{V}_{\text{ord}} \cup \{uw\}$ 
12        $S \leftarrow S \cup \text{Eq}(uw)$ 
13   $\forall uw \in \mathcal{V}_{\text{ord}}: \text{Eq}(uw) = \bigcup_{u', w' \in \mathcal{V}_{\text{ord}}} \text{Eq}(u'w')$ 
  /* Synchronize with the symmetrical order to  $\mathcal{V}_{\text{ord}}$  */
14   $\forall uw \in \mathcal{V}_{\text{ord}}: \text{Eq}(wu) = \{xv, vx \in \text{Eq}(uw)\}$ 
  /* we have equivalent classes that can induce a weak
    hanani-tutte drawing */
15  forall  $v$  in vertices of level  $h$  do
16     $G' \rightarrow G' \cup \{(w, v) \mid w \in V, (w, v) \in E \text{ and } l(w) = l(v) + 1\}$ 
17    if  $v$  is a cut vertex of  $G'$  then
18      Comps list of connected comps of  $G' - \{v\}$ 
19      we pick an arbitrary order of the list Comps
20      forall  $(C_1, C_2)$  in combinaisons of pairs of connected components in
        Comps with the respect to the specified order do
21        if  $C_1$  is encapsulated by  $C_2$  then  $(C_1, C_2) \leftarrow (C_2, C_1)$ 
22        /* we consider the vertices of the  $h$  level only */
         $\forall v \in C_1(h) \forall w \in C_2(h) : \text{Eq}(v, w) \leftarrow \bigcup_{w' \in C_2(h)} \text{Eq}(v, w')$ 

```

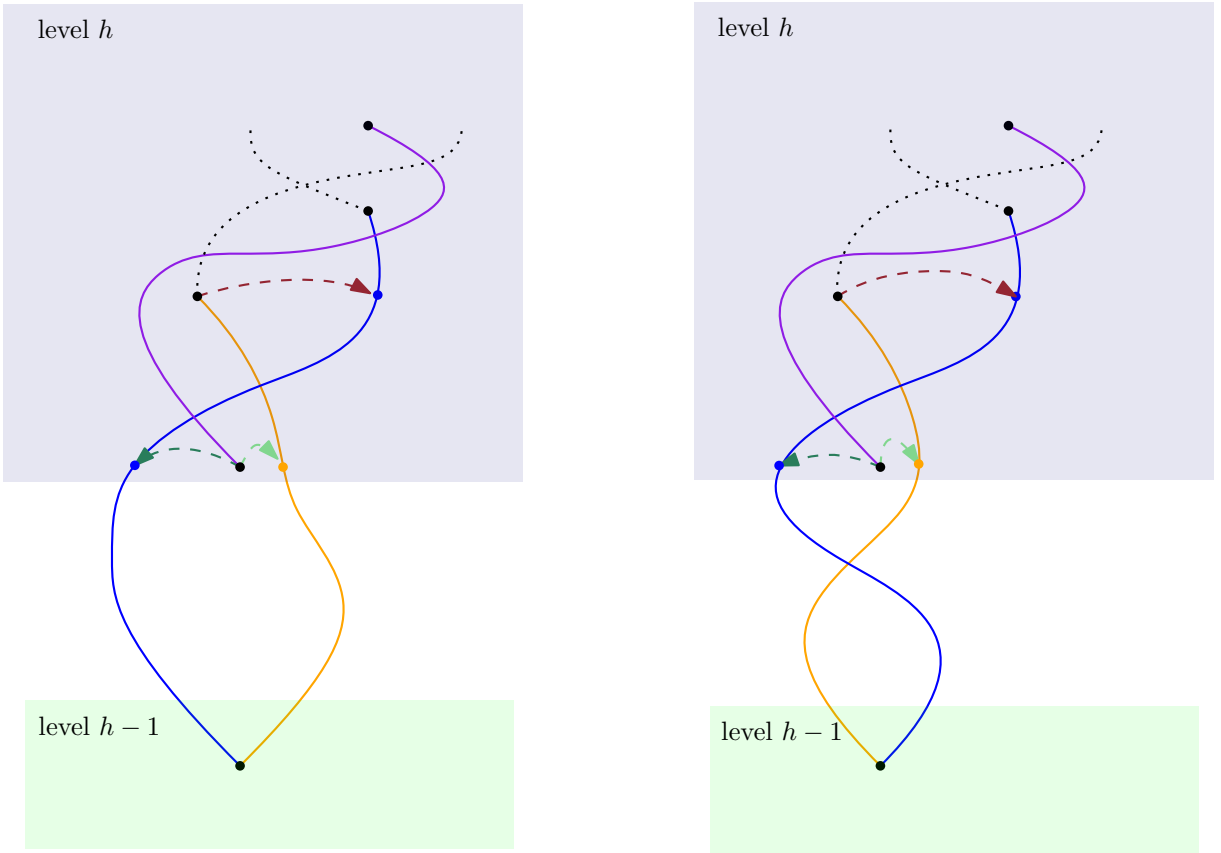


Figure 3.1.: consecutive edges in rotation that crosses oddly

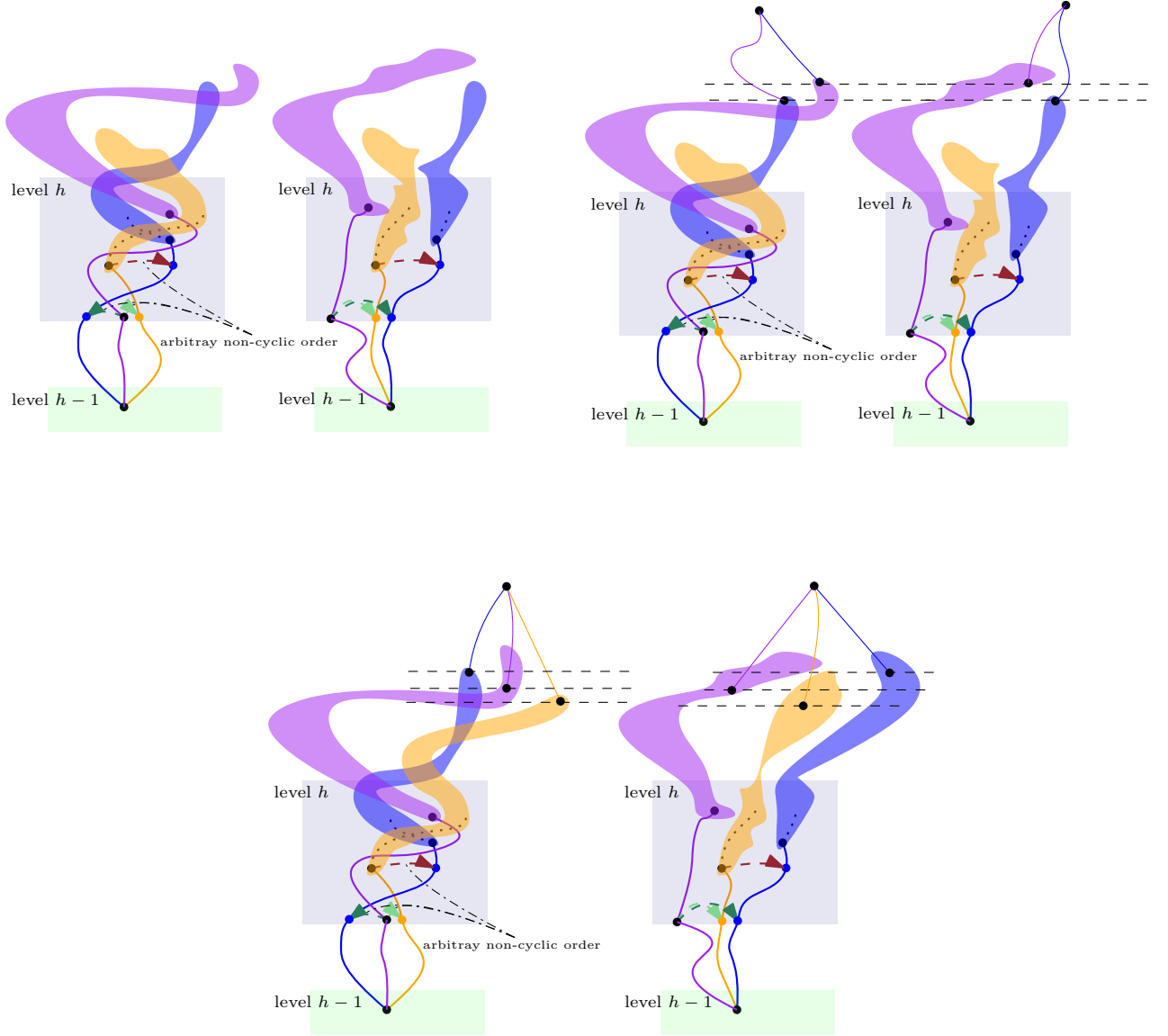


Figure 3.2.: in between even crossing edge

4. Conclusion

Summary and outlook.

Bibliography

- [BRS22] Guido Brückner, Ignaz Rutter, and Peter Stumpf. Level-planarity: Transitivity vs. even crossings. *The Electronic Journal of Combinatorics*, pages P4–22, 2022.
- [FPSŠ12] Radoslav Fulek, Michael J Pelsmajer, Marcus Schaefer, and Daniel Štefankovič. Hanani–tutte, monotone drawings, and level-planarity. In *Thirty essays on geometric graph theory*, pages 263–287. Springer, 2012.
- [Kro67] Melven R Krom. The decision problem for a class of first-order formulas in which all disjunctions are binary. *Zeitschrift für mathematische Logik und Grundlagen der Mathematik*, 13(1-2), 1967.
- [RSB⁺01] Bert Randerath, Ewald Speckenmeyer, Endre Boros, Peter Hammer, Alex Kogan, Kazuhisa Makino, Bruno Simeone, and Ondrej Cepek. A satisfiability formulation of problems on level graphs. *Electronic Notes in Discrete Mathematics*, 9:269–277, 2001.

Appendix

A. Appendix Section 1

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Figure A.1.: A figure